3 Yr. Degree/4 Yr. Honours 5th Semester Examination, 2025 (CCFUP)

Subject : Computer Science

Course : COMP5012 (MAJOR)

(Computer Organization)

Time: 2 Hours

Full Marks: 40

The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable.

1. Answer *any five* of the following questions: 2×5=10

   (a) What is an interrupt?

   (b) Define opcode.

   (c) What is an instruction cycle?

   (d) What is subroutine?

   (e) Define microprogramming.

   (f) Convert Gray Code 1011 into binary.

   (g) What is Program Counter (PC)?

   (h) What is handshake communication?

2. Answer *any two* of the following questions: 5×2=10

   (a) Differentiate between Horizontal and Vertical Micro-instruction formats.

   (b) What is the "Stored program" concept of computer architecture? Cite the differences between Von Neumann and Harvard architectures. 2+3

   (c) What is instruction pipelining? Explain its stages. 1+4

   (d) Explain Bus Organization in a computer system.

3. Answer *any two* of the following questions: 10×2=20

   (a) Explain Direct Memory Access (DMA) with block diagram and advantages. 8+2

   (b) What are Pipeline Hazards? Explain types and solutions. 2+8

   (c) Describe instruction format and addressing modes in detail. 5+5

   (d) Discuss the interrupt handling mechanism. Mention different types of interrupts. 7+3

Mem bus bottleneck
Ex Hazard